

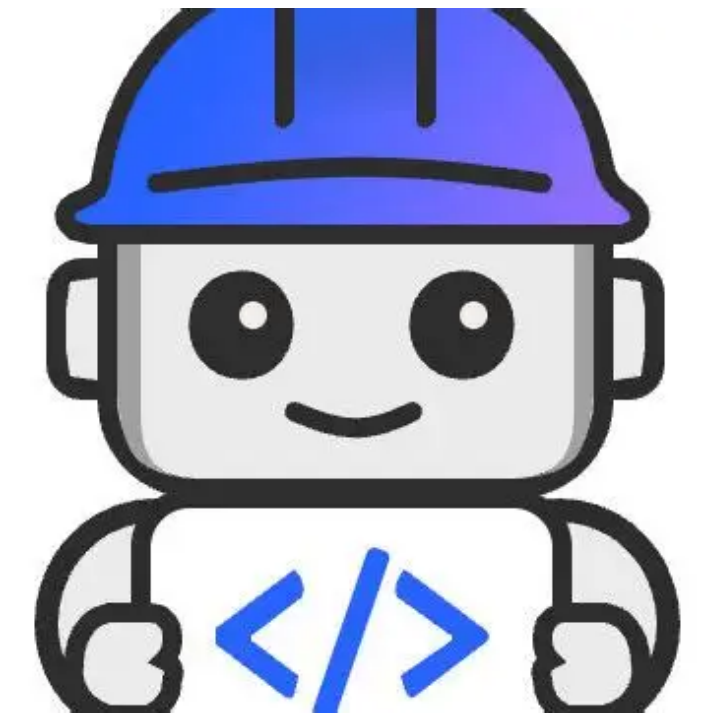
AI — IBM Bob Hackathon

SAFETYREADY

Pharmacovigilance Signal Detection &
ICH CTD Submission Readiness

Built with IBM Bob

A fully offline drug-safety platform —
developed end-to-end using AI-assisted
engineering



TEAM PARALLAXA

[GitHub](#)

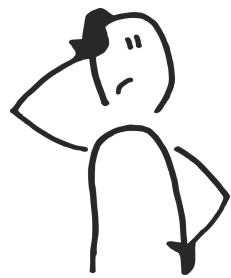
Shaikh Saniya Khatun

Vidhi Shah

Darshan Solanki

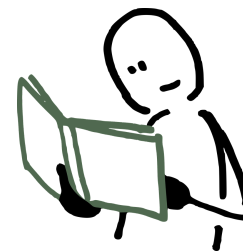
Khushi Patel

PROJECT OVERVIEW



PROBLEM

- Thousands of adverse-event reports make safety-signal detection slow and error-prone.
- Manual ICH CTD checks make regulatory gaps easy to miss.
- Result: missed signals, submission delays, and costly rejections.



WHO HAS IT

Pharmaceutical regulatory analysts and drug-safety scientists



WHY IT MATTERS

Delayed or failed submissions cost millions and — more critically — delay patient access to safe medicines.

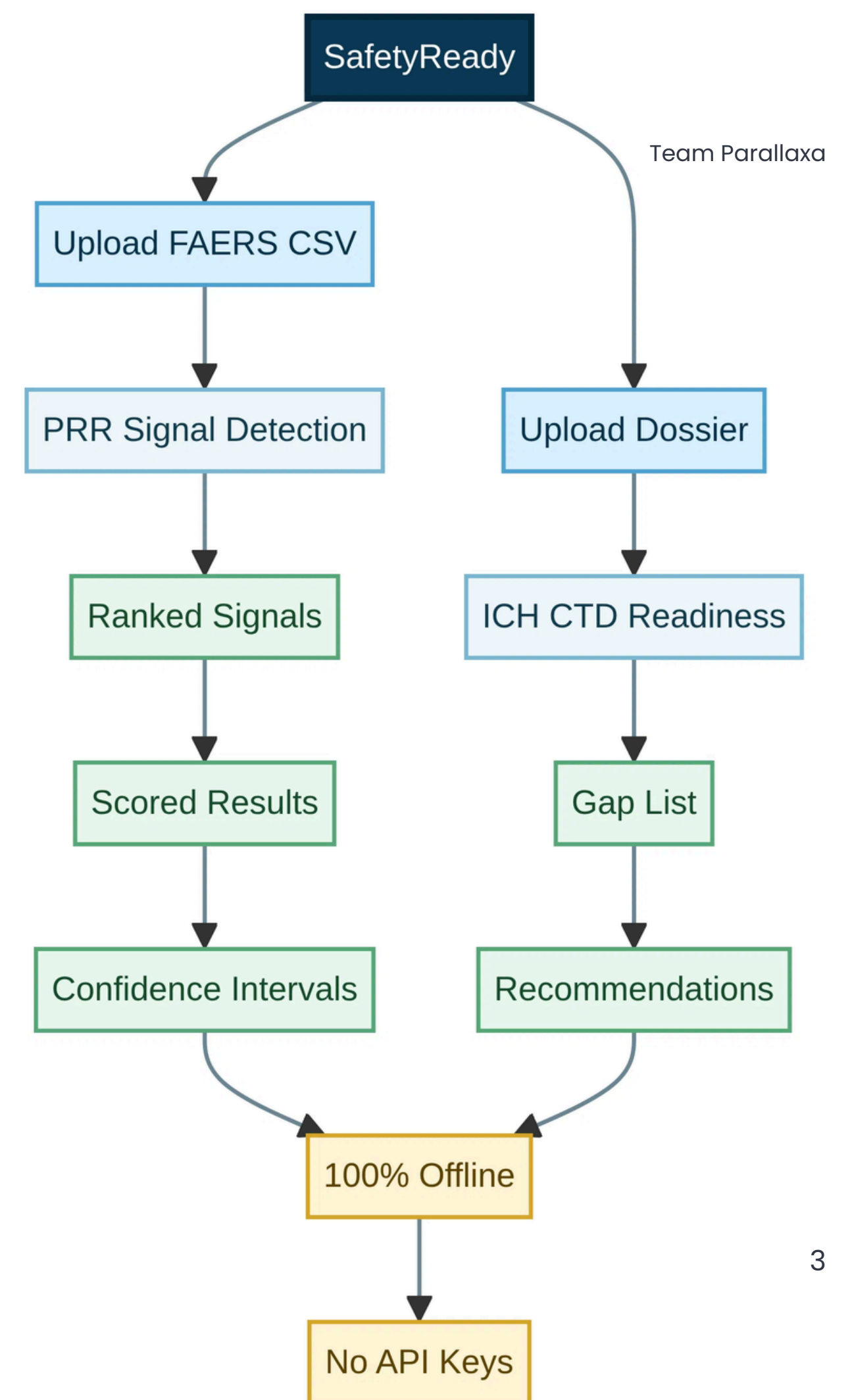
SOLUTION

FEATURE

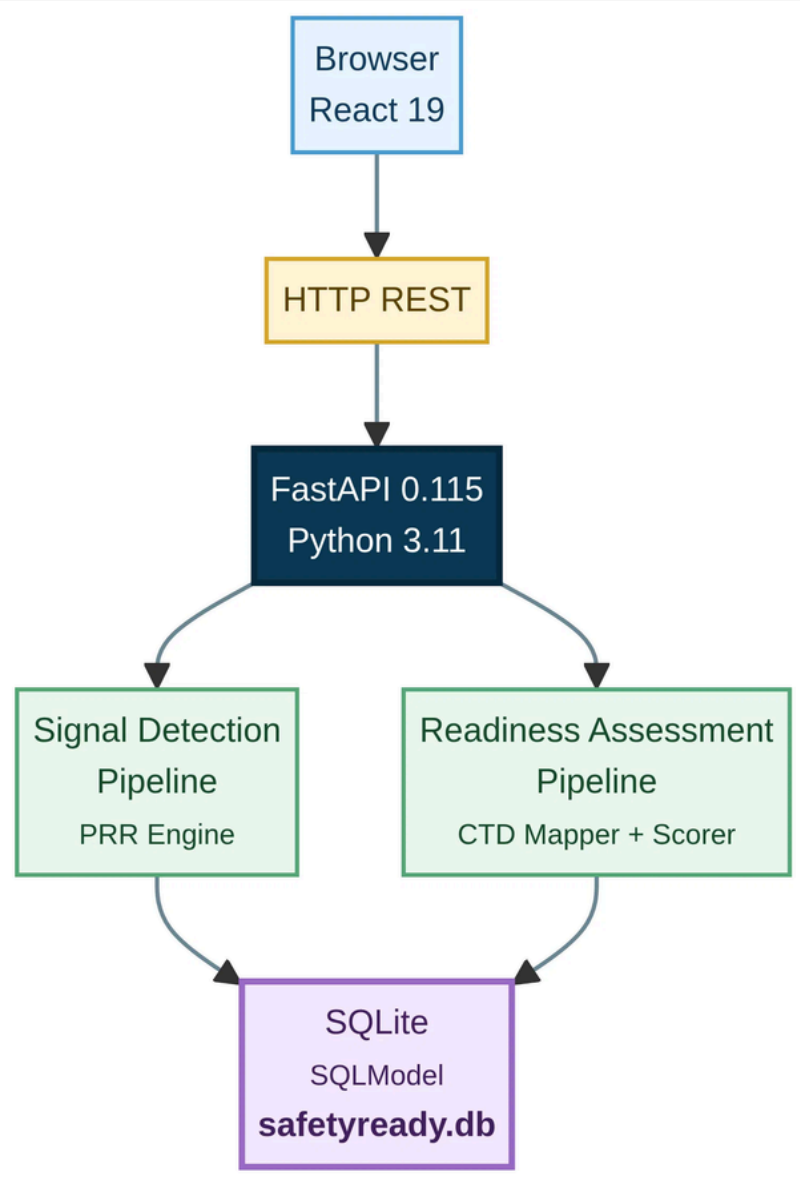
Signal Detection: PRR pipeline → ranked drug-safety signals with 95% confidence intervals

Readiness Assessment: 4-tier CTD mapping → per-module scores + prioritised gap list

One upload, live polling, instant results this could be in little detail not too big



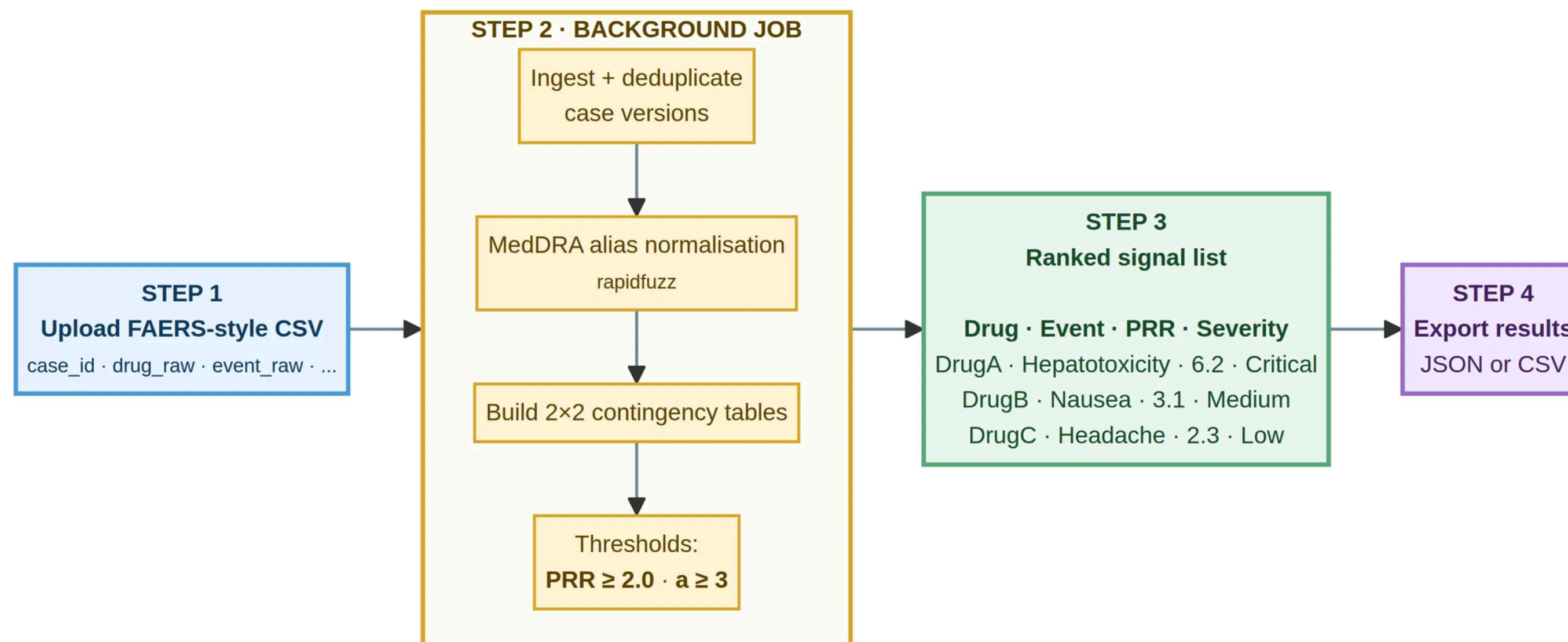
ARCHITECTURE



Layer	Technology
Frontend	React · TypeScript · Vite
Backend	FastAPI · Python · Pydantic
Core Logic	PRR Engine · CTD Mapper · rapidfuzz
Database	SQLite · SQLAlchemy
Job Processing	FastAPI BackgroundTasks
Development	IBM Bob

No external services. No cloud. No AI APIs at runtime.

DEMO / KEY FEATURE

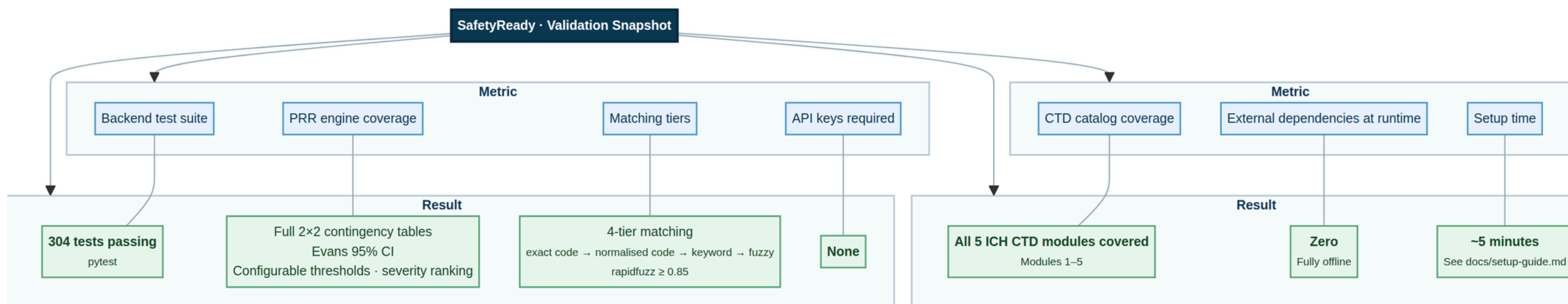


IBM TECHNOLOGIES

Bob Capability	Our Use
Code Generation	Built the core Signal Detection & Readiness engines
Test Authoring	Created 304 backend + 51 frontend tests
Type Consistency	Kept Python and TypeScript types synchronized
Documentation	Generated and refined technical documentation
Refactoring	Restructured code while keeping tests passing

IBM Bob supported the entire development process, from coding and testing to documentation and refactoring.

RESULTS



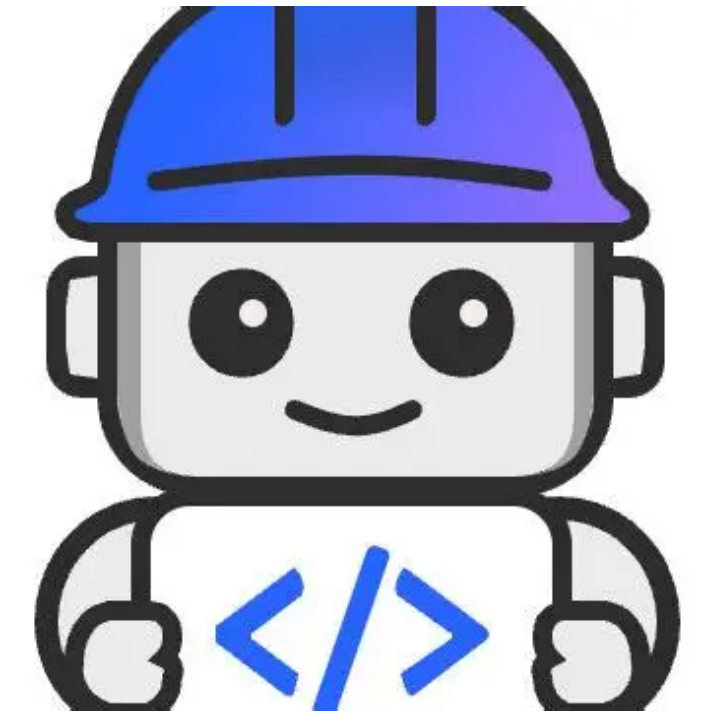
What a production version enables: Regulatory teams cut manual signal-review time from days to minutes and catch CTD dossier gaps before submission — reducing late-stage rejection risk.

TEAM

Name	Role	Contribution
Saniya Khatun Shaikh	Project Lead & Full-Stack Developer	Built the complete platform and core functionality
Vidhi Shah	Presentation & Media	PPT design and demo video editing
Darshan Solanki	Research & Coordination	Research and submission support
Khushi Patel	Demo & Documentation Support	Demo preparation and supporting materials

AI — IBM Bob Hackathon

THE END



TEAM PARALLAXA

Shaikh Saniya Khatun

Vidhi Shah

Darshan Solanki

Khushi Patel